

Multifunctional ytterbium oxide buffer for perovskite solar cells

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Perovskite solar cells (PSCs) comprise a solid perovskite absorber sandwiched between several layers of different charge-selective materials, ensuring unidirectional current flow and high voltage output of the devices^{1,2}. A ‘buffer material’ between the electron-selective layer and the metal electrode in p-type/intrinsic/n-type (p-i-n) PSCs (also known as inverted PSCs) enables electrons to flow from the electron-selective layer to the electrode^{3–5}. Furthermore, it acts as a barrier inhibiting the inter-diffusion of harmful species into or degradation products out of the perovskite absorber^{6–8}. Thus far, evaporable organic molecules^{9,10} and atomic-layer-deposited metal oxides^{11,12} have been successful, but each has specific imperfections. Here we report a chemically stable and multifunctional buffer material, ytterbium oxide (YbO_x), for p-i-n PSCs by scalable thermal evaporation deposition. We used this YbO_x buffer in the p-i-n PSCs with a narrow-bandgap perovskite absorber, yielding a certified power conversion efficiency of more than 25%. We also demonstrate the broad applicability of YbO_x in enabling highly efficient PSCs from various types of perovskite absorber layer, delivering state-of-the-art efficiencies of 20.1% for the wide-bandgap perovskite absorber and 22.1% for the mid-bandgap perovskite absorber, respectively. Moreover, when subjected to ISOS-L-3 accelerated ageing, encapsulated devices with YbO_x exhibit markedly enhanced device stability.

The power conversion efficiency (PCE) of single-junction p-type/intrinsic/n-type (p-i-n) perovskite solar cells PSCs (also referred to as inverted PSCs) has surpassed 25% through passivation methodologies^{3–5,13–15}. However, the published operational stability of p-i-n PSCs under stringent test conditions still lags behind those of commercially viable solar panels^{16–18}. Although multiple factors are responsible for device degradation under operating conditions^{19–21}, ion migration under realistic operational conditions represents an important challenge as it is almost ubiquitous within a device. Electrical bias and illumination-induced ion migration within a perovskite absorber layer can vary the chemical composition at the nano-scale and micro-scale levels^{19,21,22}, inducing higher chemical heterogeneity and increased non-radiative recombination centres and subsequent initial sites at which device degradation occurs^{2,23}. Furthermore, these mobile ions from the perovskite absorber can diffuse into the charge-selective layers and may accumulate at the metal electrode interface. The accumulated ions corrode the metal electrodes^{7,24}, damaging the integrity of

the metal electrode and degrading the stability of the device. Another key concern for long-term instability is the metal diffusion into the electron-selective and perovskite layers, which may lead to the formation of metal halides and weakened device stability^{7,25,26}.

Attempting to introduce an effective buffer layer between the electron-selective layer (ESL) and the metal electrode to block atom and ion movement between the perovskite and metal layers (in both directions; Extended Data Fig. 1a) is of technological interest in improving overall device performance^{6–8}. In this case, the buffer layer must facilitate the directional flow of electrons from the buffer to the metal electrode, ensuring efficient electron collection of the PSCs. The method for depositing the buffer layer must be compatible with underlying materials such as (6,6)-phenyl C₆₁ butyric acid methyl ester and buckminsterfullerene (C₆₀). Moreover, an ideal buffer layer can effectively prevent the permeation of external harmful species such as O₂ and moisture from the air. To date, bathocuproine (BCP), the most commonly used buffer material, can be processed with scalable

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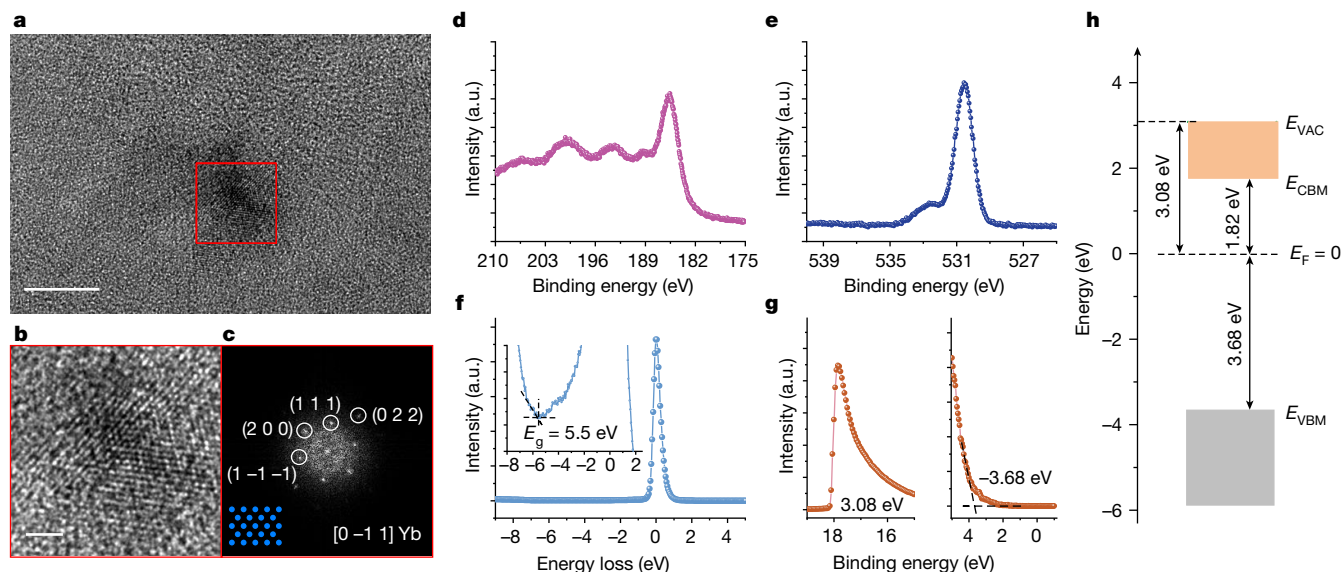


Fig. 1 | Characterization of YbO_x films. **a**, A general HRTEM image of the pure Yb film. **b, c**, The magnified HRTEM image (**b**) and the corresponding FFT patterns (**c**) of the red squared area selected in **a**. **d, e**, XPS spectra of the Yb 4d corresponding to the Yb^{3+} state (**d**) and O 1s (**e**). **f**, REELS spectrum of the YbO_x film. The inset shows how the band gap is determined. **g**, UPS spectra of the

YbO_x film. Left, secondary electron cut-off spectrum. Right, valence band spectrum. **h**, Energy levels of the YbO_x film. Key parameters shown are E_{CBM} (conduction band minimum), E_{VBM} (valence band maximum), E_{F} (Fermi level), and E_{VAC} (vacuum level). Scale bars, 10 nm (**a**); 2 nm (**b**). a.u., arbitrary unit.

thermal evaporation in p-i-n PSCs⁵⁹. Unfortunately, BCP suffers from poor thermal stability under the protocol temperature of 85 °C by the International Summit on Organic Photovoltaic Stability (ISOS) protocols, which will probably limit long-term operational stability^{12,22,27–29}. Low work-function oxides, including zinc oxide (ZnO) nanoparticles, compact tin oxide (SnO_2) layer and aluminium-doped zinc oxide nanoparticles, have also been used as buffer materials in p-i-n PSCs to replace the BCP or to form bilayer buffers^{11,12}. Yet, these oxides either are challenging for large-scale fabrication or require dedicated instruments and expensive fabrication precursors, which contradicts cost-effective practical applications. An intrinsically stable and multifunctional material, ytterbium oxide (YbO_x), which can be deposited by the scalable thermal evaporation approach, is a compelling buffer material for efficient and stable p-i-n PSCs.

Properties of ytterbium oxide

Ytterbium (Yb) with a low Gibbs free energy has a strong tendency to form YbO_x when exposed to O_2 (Extended Data Fig. 1b and Supplementary Note 1). The YbO_x that can be processed with thermal evaporation (Extended Data Fig. 1c) shows great promise in commercial applications (Supplementary Note 2 and Supplementary Tables 1 and 2). To understand the oxidation of Yb, we first thermally evaporated a few nanometres of Yb films on a large-scale indium tin oxide (ITO) substrate (100 cm²) in the device fabrication vacuum chamber with a base pressure of 10^{-4} Pa. The sampling data from different areas of the sample shows identical YbO_x states (Extended Data Fig. 2). This rapid oxidation associated with metal Yb is further demonstrated by the colour change of the polished Yb metal surfaces before and after oxidation (Extended Data Fig. 3). We then studied the detailed oxidation dynamics of the clean Yb^0 surface by tracking the oxidation process over time using in situ X-ray photoelectron spectroscopy (XPS) (Extended Data Fig. 4 and Supplementary Table 3). These results indicate that the Yb film (with a thickness of a few nanometres) deposited on the large-area ITO substrate can be fully oxidized in a few minutes when upscaling in a device fabrication vacuum chamber with a base pressure of 10^{-4} Pa (see details in Supplementary Note 3). These benefits eliminate the additionally dedicated oxidation step,

which is usually required for other oxides processed with thermal evaporation³⁰.

Figure 1a–c shows the high-resolution transmission electron microscopy (HRTEM) images and the corresponding fast Fourier transform (FFT) patterns of the Yb film. The HRTEM images and FFT patterns feature the crystalline structure corresponding to a face-centred cubic system with space group 225 (Fm3m) visualized from the [0 –1 1] direction in the selected red squared region. By clarifying the Yb natural oxidation through XPS spectra^{31,32} (Fig. 1d,e), the resulting YbO_x exhibits a broad and continuous halo in the typical FFT pattern, indicative of the amorphous atomic arrangement³³ (Extended Data Fig. 5). The amorphous characteristics of the YbO_x films are also confirmed through grazing-incidence wide-angle X-ray scattering (Extended Data Fig. 6). The energy-dispersive X-ray chemical composition map for the YbO_x film suggests that a distribution of both Yb and O elements in the films is spatially homogeneous (Supplementary Fig. 1). The reflection electron energy loss spectroscopy (REELS) spectrum shows that the surface bandgap of the YbO_x is as high as 5.5 eV (ref. 34) (Fig. 1f), which is transparent to visible light wavelengths. Important information associated with the electronic structure, including work function, valence band maximum (VBM) and conduction band minimum, is quantified by combining ultraviolet photoelectron spectroscopy (UPS) analyses (Fig. 1g) with the REELS data. These results indicate that the YbO_x film may be n-type, with a Fermi level of 3.08 eV from the vacuum and a conduction band minimum of 1.26 eV from the vacuum (Fig. 1h).

Multifunctional role of interfacial YbO_x

Figure 2 shows XPS data on the variation of chemical constituents in the ESL and its interface with the metal electrode measured from samples with (target) and without (control) the YbO_x buffer layer before and after accelerated ageing in ambient air (relative humidity, 80%) at 80 °C for 5 h. The samples consist of substrate/perovskite/ C_{60} /Cu (control) and substrate/perovskite/ C_{60} / YbO_x /Cu (target). A mechanical lift-off approach was used to delaminate the Cu electrode from the rest of the sample stacks³⁵. The delaminated Cu is referred to as the Cu side, and the other is the substrate side (Supplementary Fig. 2). Supplementary Fig. 3 shows scanning electron microscope images of the delaminated

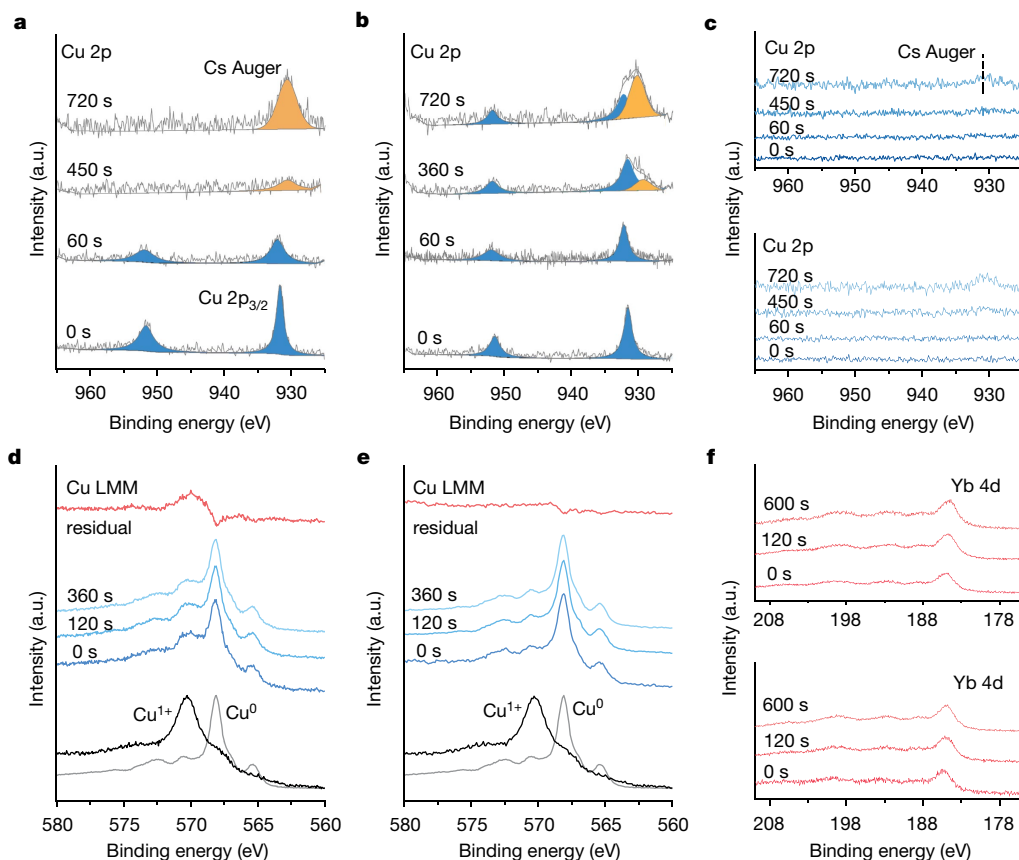


Fig. 2 | Photoelectron spectroscopy of the lift-off full devices. a, b, Cu 2p XPS depth profiling using GCIB sputtering (the sputtering rate for organic films is about 0.05 nm s^{-1} using *N,N'*-bis-(1-naphthyl)-*N,N'*-diphenyl-1,1'-biphenyl-4,4'-diamine as the reference) detected at the substrate side of the fresh control (without YbO_x) sample (a) and the aged control sample (b) (after thermal annealing at 80°C in ambient air for 5 h). Control: substrate/perovskite/ C_{60} /Cu. c, Cu 2p XPS depth profiling detected at the substrate side of the fresh (top) and the aged (bottom) target samples (with YbO_x , after thermal annealing at 80°C in ambient air for 5 h). Target: substrate/perovskite/ C_{60} / YbO_x /Cu. d, e, Cu LMM

Auger spectra were detected at the Cu side of (d) the aged control sample and (e) the aged target sample (after thermal annealing at 80°C in ambient air for 5 h). The red line at the top shows the spectra difference, and this line is obtained by using the as-delaminated spectrum (0 s, blue lines at the middle) minus Cu^0 reference spectrum (grey lines at the bottom), which can be used to analyse chemical states. f, Yb 4d XPS depth profiling measured at the Cu side of the target sample with YbO_x , before (top) and after (bottom) the accelerated ageing. a.u., arbitrary unit.

surfaces of both control and target samples. The buried organic C_{60} layer for the substrate side was removed using gas cluster ion beam (GCIB) sputtering to probe throughout the layer stack.

Figure 2a, b shows the Cu 2p XPS depth profiling of the substrate side before and after ageing. Note that the Cs Auger signal (highlighted in orange) is used to probe the perovskite layer in this observation. The data show a noticeable decrease in the Cu 2p XPS peak intensity of the fresh control sample (without YbO_x) after the 60 s of GCIB sputtering, and the Cu 2p XPS peak intensity is below the limitation of the instrument with increasing the sputtering time. By contrast, the Cu 2p XPS peaks for the aged control samples remain detectable after extended GCIB sputtering, for example, to 360 s (Fig. 2b). Whereas at 720 s, Cu 2p XPS peaks and notable Cs Auger signals are simultaneously detected for the aged control sample. This experimental observation indicates that Cu diffusion exists in the aged control samples. When the YbO_x buffer layer is inserted between the C_{60} and Cu, the distribution of the Cu species into the C_{60} and perovskite underneath is not detected for both fresh and aged target samples (Fig. 2c). Next, we probed the chemical changes on the delaminated Cu surfaces of both control and target samples by tracking the Cu LMM Auger peaks of the delaminated Cu side after ageing. A peak exists at around 571 eV for the aged control sample without YbO_x (Fig. 2d). This result indicates that Cu oxidation can happen in the aged control samples, consistent with the colour change of the Cu

electrode (control) after accelerated ageing in ambient air (relative humidity, 80%) at 80°C for 5 h (Supplementary Fig. 4). However, the Cu oxidation peak is not observed in the aged target samples (with YbO_x)^{23,36} (Fig. 2e). The depth-profiling data show that the chemical state of the YbO_x remains the same throughout the thickness before and after accelerated ageing (Fig. 2f).

The UPS depth profiling analysis based on the real device stacks shows that the insertion of YbO_x can reduce the work function of the contact interface compared with the control sample without YbO_x (Supplementary Fig. 5). One such variation brings a positive impact on electron collection at the Cu electrode interface. To show the charge-carrier transport properties of the YbO_x buffer, two devices with 20 alternative layers of YbO_x (2 nm) and YbO_x (5 nm)/Cu (1 nm) (for example, $\text{ITO}/[\text{YbO}_x(2 \text{ nm})]_{20}/\text{Cu}(100 \text{ nm})$ and $\text{ITO}/[\text{YbO}_x(5 \text{ nm})/\text{Cu}(1 \text{ nm})]_{20}/\text{Cu}(100 \text{ nm})$) were prepared. We confirm that the YbO_x used in the charge-carrier transport measurements shows no metallic states and instead exhibits a considerable density of in-gap states near the Fermi level (Extended Data Fig. 7a, b). By measuring the temperature dependence of current density–voltage (J – V) characteristics of the devices with $\text{ITO}/[\text{YbO}_x(2 \text{ nm})]_{20}/\text{Cu}(100 \text{ nm})$ and $\text{ITO}/[\text{YbO}_x(5 \text{ nm})/\text{Cu}(1 \text{ nm})]_{20}/\text{Cu}(100 \text{ nm})$ (Extended Data Figs. 7c, d and 8), we find that the charge-carrier transport inside the YbO_x follows a thermally activated hopping from one localized state to another rather than through the drift of Bloch wave described in the typical energy-band

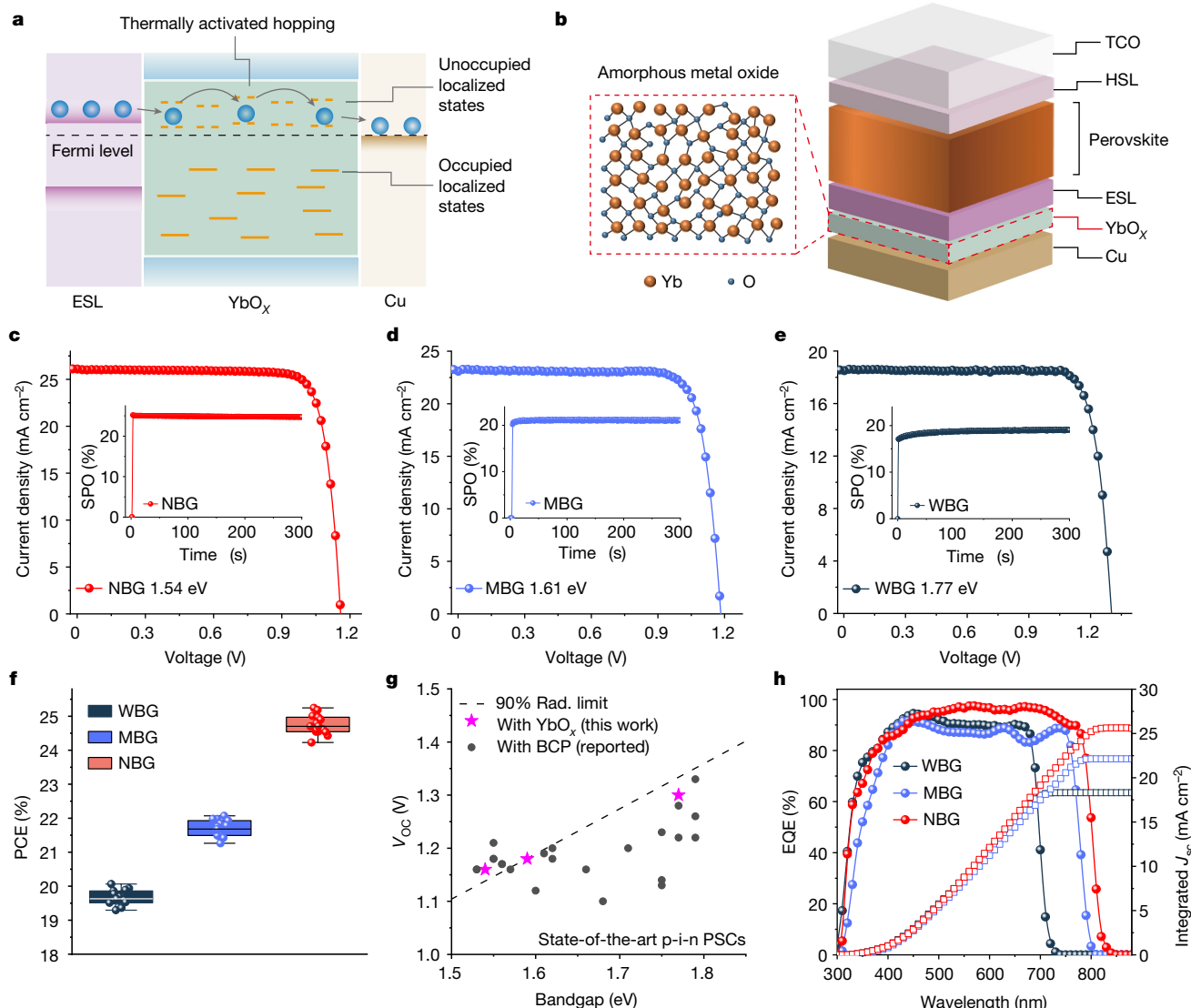


Fig. 3 | Photovoltaic performance. **a**, Schematic of the charge-carrier transport across the ESL/YbO_x/Cu interface. The carrier transport in the YbO_x buffer layer is by thermally activated hopping from one localized state to another. **b**, Schematic of the device configuration of a typical p-i-n PSCs with the YbO_x buffer layer. The magnified area illustrates the amorphous nature of the YbO_x buffer layer used in our p-i-n PSCs. ESL, electron-selective layer; TCO, transparent conducting oxide. **c–e**, The *J*–*V* characteristics of the p-i-n PSCs with various bandgaps and SPO of the devices at the maximum power point under simulated

AM1.5 G illumination (inset). NBG (**c**), MBG (**d**) and WBG (**e**). **f**, Distribution of PCE for p-i-n PSCs with NBG, MBG and WBG. The number of devices is 15 for each. **g**, Comparison of the *V*_{oc} of state-of-the-art p-i-n PSCs with varied bandgaps. Note that the data from the literature is based on the p-i-n PSCs with the BCP buffer layer. The dashed line represents 90% of the radiative limit concerning the bandgap energy. **h**, The EQE spectra of the champion p-i-n PSCs with NBG, MBG and WBG perovskites.

theory (Supplementary Note 4). This physical process was originally proposed by Mott³⁷, who gave the temperature–conductivity dependence law below:

$$\sigma = \sigma_0 \exp\left(-B/T^{1/4}\right) \quad (1)$$

where σ_0 is a pre-factor and B depends on the radial extension of the wave functions and density of states. Herein, we illustrate phonon-assisted electron hopping by unoccupied localized states of the YbO_x buffer layer when electrons flow from Cu₆₀ to Cu (Fig. 3a).

Photovoltaic performance enhancement

To assess the impact of the YbO_x buffer layer on device performance, we fabricated full devices using perovskites with three different

typical bandgaps. The cross-sectional and top-view scanning electron microscope images for these perovskites are shown in Supplementary Fig. 6. Supplementary Fig. 7 shows the optoelectronic properties of the associated perovskites with various bandgaps. First, we evaluated the photovoltaic performances of the p-i-n PSCs based on a narrow-bandgap (NBG) perovskite absorber (Rb_{0.05}CS_{0.05}MA_{0.05}FA_{0.85}Pb(I_{0.95}Br_{0.05})₃). Then, the perovskite absorber was post-treated with 3-(aminomethyl)pyridine to suppress non-radiative recombination at the interface³. Figure 3b shows device configurations of the p-i-n PSCs with YbO_x and the amorphous nature of the YbO_x buffer. The optimal thickness for the YbO_x layer in this work is around 1.5 nm (Supplementary Fig. 8).

From the *J*–*V* scanned results, the champion PSC with YbO_x exhibits a remarkable PCE of 25.2% (an open-circuit voltage (*V*_{oc}) of 1.16 V, a fill factor (FF) of 0.83 and a short-circuit current density (*J*_{sc}) of 26.1 mA cm⁻²) (Fig. 3c and Extended Data Table 1). The steady-state

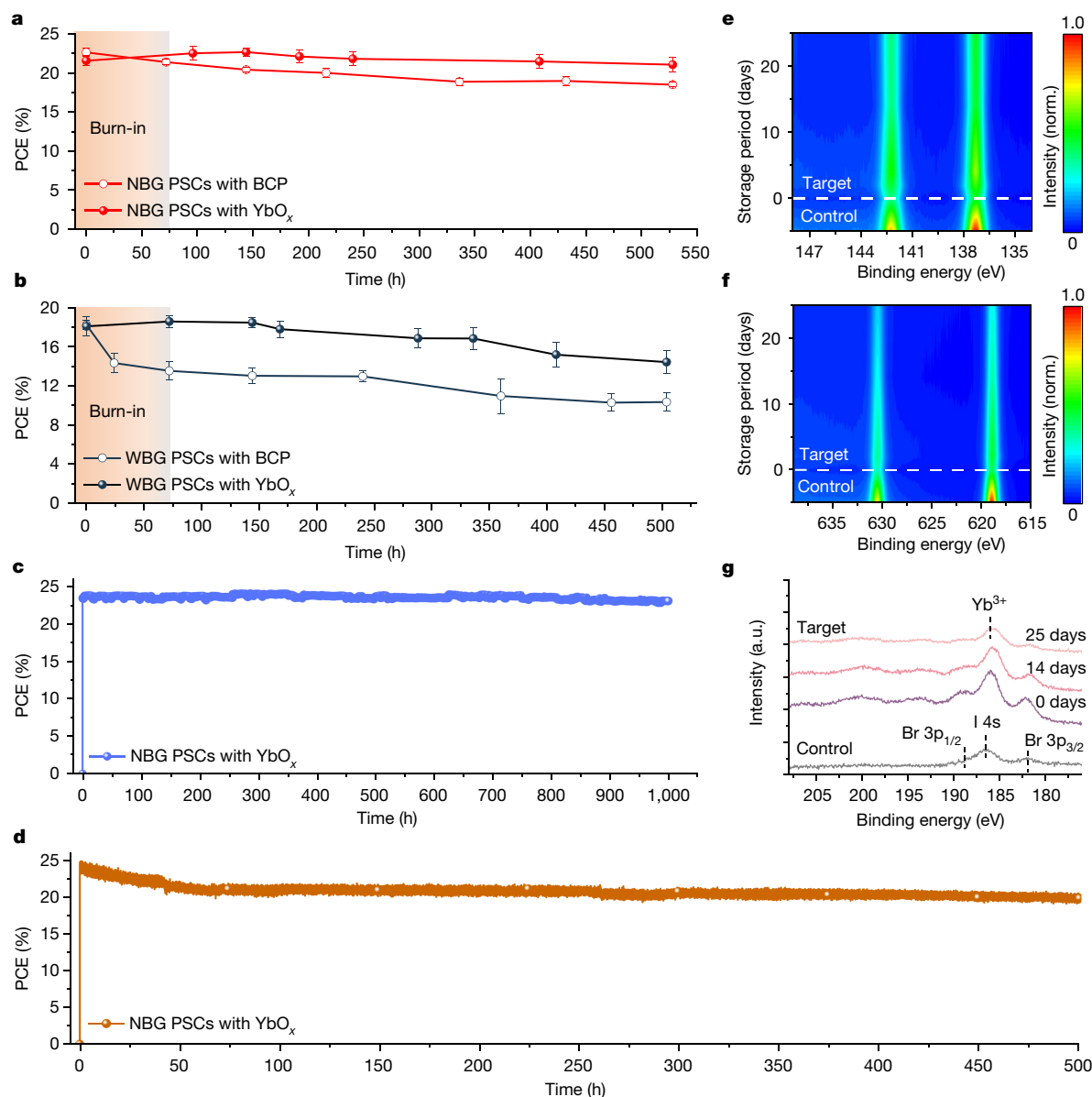


Fig. 4 | Stability of YbO_x-based p-i-n PSCs. **a, b**, Variations of the PCEs of the encapsulated NBG-containing PSCs (**a**) and WBG-containing PSCs (**b**) with the BCP or YbO_x buffer layer aged in an N₂-filled glove box under 85 °C in the dark. **c**, The maximum power point tracking (MPPT) curve of an encapsulated NBG-containing PSC with YbO_x under continuous illumination by using a white LED array with an intensity of 100 mW cm⁻² in an N₂ atmosphere at about 40 °C. **d**, The MPPT curve of an encapsulated NBG-containing PSC with YbO_x held at 85 °C under 1-sun illumination and 50% relative humidity in ambient air (ISOS-L-3

conditions). The light source is a white light LED solar simulator with a simulated AM1.5 G spectrum. The temperature was controlled and monitored by polyimide heating film with a digital-displayed temperature controller. **e–g**, Evolutions of Pb 4f (**e**), I 3d (**f**) and Yb 4d (**g**) XPS peaks of the perovskite/YbO_x bilayers as a function of the storage period in an N₂-filled glovebox. The perovskite film without the YbO_x is marked as 'control' and the perovskite/YbO_x bilayer is marked as 'target'. a.u., arbitrary unit; norm., normalized.

power output (SPO) of our YbO_x-based p-i-n PSCs is about 24.7% under standard simulated 1-sun illumination, AM1.5 G irradiance (Fig. 3c) and the YbO_x-based devices show negligible hysteresis in nature (Supplementary Fig. 9). Note that a MgF₂ anti-reflection coating was used to minimize the light reflection to maximize the photocurrent of the p-i-n PSCs (Supplementary Table 4). To further validate our finding, we sent one of our non-encapsulated YbO_x-based NBG PSCs to an accredited institution of the Photovoltaic and Wind Power Systems Quality Test Center (Chinese Academy of Sciences) for external certification. We obtained a certified PCE of 25.09% (Supplementary Fig. 10), which is among the highest PCEs of p-i-n PSCs and represents the one of the best PCEs of p-i-n PSCs with an inorganic buffer layer (Supplementary Tables 5 and 6). Our experiments further validate the universality of

the YbO_x buffer layer in the p-i-n PSCs with mid-bandgap (MBG) and wide-bandgap (WBG) perovskite absorbers. The MBG PSCs with YbO_x deliver notable performance with a champion PCE of 22.1% (Fig. 3d). In particular, the YbO_x-based WBG devices yield an impressive PCE of up to 20.1% (a V_{oc} of 1.30 V, an FF of 0.82 and a J_{sc} of 18.6 mA cm⁻²) (Fig. 3e). To our knowledge, this PCE is among the highest PCEs for p-i-n PSCs with WBGs of 1.75–1.80 eV (Supplementary Table 7). Moreover, we find that p-i-n PSCs with YbO_x show high reproducibility and minor batch-to-batch variations in device performance regardless of the perovskite bandgap (Fig. 3f and Supplementary Figs. 11, 12).

Figure 3g shows that all these p-i-n PSCs have achieved prominent V_{oc} values, which are close to or even surpass 90% of the radiative limit versus the bandgap energy³⁸. Furthermore, external quantum

efficiencies of the electroluminescence (EQE_{EL} s) for the YbO_x -based p-i-n PSCs with NBG and WBG perovskite absorbers (Supplementary Fig. 13) present 5.12% (NBG) and 0.97% (WBG), respectively. Under these conditions, derived from the EQE_{EL} , we calculated the QFLS values of the PSCs containing the different absorbers to be 1.159 eV (NBG) and 1.289 eV (WBG), respectively, consistent with the obtained V_{OC} from the J - V scans (Supplementary Note 5). Moreover, the integrated J_{SC} values of the champion NBG-, MBG-, and WBG-based PSCs, in which their bandgaps are calculated to be 1.54 eV, 1.59 eV, and 1.77 eV (Supplementary Fig. 14), are 25.7 mA cm^{-2} , 22.1 mA cm^{-2} , and 18.3 mA cm^{-2} , respectively (Fig. 3h).

Thermal and operational stability

We tested the thermal stability of the PSCs with BCP or YbO_x (encapsulated using glass covers sealed by epoxy resin). The encapsulated PSCs are stored at 85°C in the dark in an N_2 -filled glovebox. Figure 4a compares the results of the thermal stability of the NBG-based PSCs with BCP or YbO_x buffer layers. Compared with the PSCs with BCP, the YbO_x -based device exhibits a slight increase in the PCEs initially (<100 h) and retains 98% of its initial PCE after 500 h. By contrast, the BCP-based device shows a sharp PCE drop on exerting thermal stress, and only 82% of its initial PCE can be retained after 500 h. Similarly, in the WBG-based PSCs, the YbO_x -based device exhibits enhanced thermal stability (Fig. 4b). However, the WBG-PSCs with either BCP or YbO_x buffer layers degrade at faster rates compared with the NBG-based PSCs. This difference correlates with the inherently poor stability of the WBG perovskite with a high bromide ratio^{39,40}. Nonetheless, the WBG-based PSCs with YbO_x can still retain 80% of their initial PCE after 500 h, whereas the BCP-based device drops quickly to 75% of its initial PCE in the initial 24 h and approximately 57% of device performance after 500 h.

We also examined the long-term operational stability tracking at the maximum power output under 1-sun illumination with and without thermal stress. The tested PSCs were also encapsulated and continuously illuminated by a white light-emitting diode (LED) array with an intensity equivalent to 100 mW cm^{-2} of AM 1.5 G irradiance without active cooling. The most stable NBG-based device with YbO_x maintained its initial performance for 800 h and retained 97% of the initial PCE (23.5%) after 1,000 h (Fig. 4c). In comparison, the NBG-based device with BCP dropped quickly to 96% of the initial PCE after 213 h (Supplementary Fig. 15). Notably, when subjected to multiple external stress (complied with ISOS-L-3 protocols), compared with the BCP-based device, the encapsulated YbO_x -based device still exhibits improved long-term operational stability tracking at maximum power output point at 85°C in ambient air (relative humidity, 50%) (Fig. 4d and Supplementary Figs. 16 and 17). Specifically, the encapsulated devices with YbO_x retain about 85% of its initial PCE (23.6%) after 500 h, which is comparable to p-i-n PSCs with SnO_2 (Supplementary Table 8). By contrast, the BCP-based device retains only 60% of the initial PCE after 213 h under ISOS-L-3 test qualifications. These results indicate that the PSCs with the YbO_x buffer show more substantial resistance to external stress than the regular BCP-containing ones, consistent with our observation above. Note that although enhanced device stability of p-i-n PSCs using BCP buffer layers in conjunction with Cr/Au electrodes has been reported in the past⁴¹, such a multilayer buffer makes the device much more complex, and Au is not a scalable metal owing to its high cost (Supplementary Table 2).

To gain more insights into the device stability improvements, we then conducted time-of-flight secondary ions mass spectrometry measurements on full PSC devices with BCP or YbO_x before and after ageing. Combining the time-of-flight secondary ions mass spectrometry with the depth-profile XPS results (Fig. 2, Extended Data Fig. 9 and Supplementary Fig. 18), we demonstrate that replacing BCP with YbO_x can suppress undesirable bi-directional harmful species movements

(such as I^- , Cs^+ , Rb^+ and Cu), and we find that the YbO_x is chemically unreactive with the perovskites (Fig. 4e–g, see details in Supplementary Note 6). Hence, we conclude that multifunctional YbO_x is advantageous for use as the buffer material from various evaluation angles (Supplementary Fig. 19).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06892-x>.

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Methods

Materials

Lead diiodide (PbI_2 , 99%), lead dibromide (PbBr_2 , 99%), (2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl)phosphonic acid (MeO-2PACz) and (2-(9H-carbazol-9-yl)ethyl)phosphonic acid (2PACz) were purchased from Tokyo Chemical Industry. Formamidinium iodide (FAI), methylammonium iodide and methylammonium bromide (MABr) were purchased from Greatcell Solar. Rubidium iodide (RbI , >99.99%) was purchased from Alfa Aesar. Caesium iodide (CsI , >99.99%) and 1-butyl-1-methylpiperidinium tetrafluoroborate ($[\text{BMP}]^+[\text{BF}_4]^-$, 99%) were purchased from Sigma Aldrich. (6,6)-phenyl- C_{61} -butyric acid methyl ester (PCBM) was purchased from Nano-C Tech. Poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) (PTAA; $M_w \approx 13,000$) and phenylethylammonium iodide (PEAI) were purchased from Xi'an Polymer Light Technology. Nickel oxide (NiO_x) nanoparticles, C_{60} and BCP were purchased from Advanced Election Technology. Ytterbium (99.9%) was purchased from Kurt Lesker (Shanghai) Trading. Magnesium fluoride (MgF_2 , 99.99%) was purchased from Alfa Aesar. Ultra-dry dimethylformamide (DMF), ultra-dry dimethyl sulfoxide (DMSO), ultra-dry isopropanol (IPA), ultra-dry ethanol, ultra-dry anisole and ultra-dry chlorobenzene were purchased from commercial sources (Acros). Toluene was purchased from Sinopharm Chemical Reagent. Cu was purchased from a commercial source with high purity. All reagents were used as received without further purification.

Perovskite precursor solution

The perovskite composition in the NBG perovskite is $\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$. The perovskite precursor solution was prepared by mixing CsI (19.5 mg), RbI (15.9 mg), MABr (8.4 mg), FAI (219.5 mg), PbI_2 (656.9 mg) and PbBr_2 (27.5 mg) in 1 ml mixed solvent of DMF:DMSO (4:1, v/v) and the resulting precursor solution was stirred at 50 °C for 2 h before use³. For the MBG perovskite, the perovskite composition is $(\text{Cs}_{0.05}\text{FA}_{0.85}\text{MA}_{0.10})_{0.95}\text{Pb}(\text{I}_{0.85}\text{Br}_{0.15})_3$. The perovskite precursor solution was prepared by mixing PbI_2 (1.28 M), FAI (1.21 M), MABr (0.16 M), PbBr_2 (0.22 M) and CsI (0.07 M) in a mixed solvent of DMF:DMSO (4:1, v/v), and the resulting precursor solution was stirred at 70 °C for 2 h before use. For the WBG perovskite, the perovskite composition is $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$. The perovskite precursor solution was prepared by mixing PbI_2 (0.56 M), FAI (1.16 M), PbBr_2 (0.84 M) and CsI (0.24 M) in a mixed solvent of DMF:DMSO (4:1, v/v), and the resulting precursor solution was stirred at 50 °C for 5 h before use.

Device fabrication

Before device fabrication, the pre-patterned ITO or FTO substrates were ultrasonically cleaned using an aqueous detergent solution, deionized water, acetone and IPA for 20 min. Subsequently, the as-cleaned ITO and FTO substrates were treated with UV-ozone for 10 min.

For the WBG perovskite, the 15 mg ml^{-1} NiO_x nanoparticle in a mixed solvent of IPA:deionized water (1:3) solution was spin-coated onto ITO substrates at 3,000 rpm for 30 s (with a ramping rate of 1,000 rpm s^{-1}), and the as-formed films were heated at 150 °C for 30 min in ambient air. The subsequent overlayers were deposited after the substrates cooled down to room temperature. The following operations were all carried out in an N_2 -filled glovebox (with H_2O and O_2 concentrations of less than 0.1 ppm). Before depositing the perovskite layer, the NiO_x film was modified by the self-assembling monolayer of 2PACz and MeO-2PACz. The solutions of 2PACz and MeO-2PACz with the same concentration (0.5 mg ml^{-1} in ethanol) were mixed with the volume ratio of 1:1, and then the mixed solution of 2PACz and MeO-2PACz was spin-coated onto the NiO_x film at 3,000 rpm (with a ramping rate of 1,000 rpm s^{-1}) for 30 s followed by annealing at 100 °C for 5 min. When the films cooled down to room temperature, the perovskite precursor solution was deposited onto the substrates by a consecutive two-step program at 1,000 rpm for 10 s with a ramping rate of 1,000 rpm s^{-1} and 5,000 rpm for 35 s

with a ramping rate of 1,000 rpm s^{-1} . During the second step, 300 μl of anisole doped with the ionic liquid ($[\text{BMP}]^+[\text{BF}_4]^-$, 0.01 mg ml^{-1}) was poured on the centre of the spinning substrates at 5 s before the end of the whole program. Then the samples were immediately transferred to the hotplate, annealing at 100 °C for 30 min. For the post-treatment of the WBG perovskite films, 200 μl of PEAi (5 mg ml^{-1} in IPA) and 100 μl of MAI (5 mg ml^{-1} in IPA) were mixed in 705 μl of a mixed solvent of DMF:IPA (5:700, v/v) to obtain 1,005 μl of post-treatment solution. Then 50 μl of the post-treatment solution was dynamically spin-coated on top of the perovskite film, followed by annealing at 100 °C for 5 min.

For the MBG perovskite, 2 mg ml^{-1} PTAA (in toluene) solution was spin-coated onto ITO substrates at 5,000 rpm for 30 s (with a ramping rate of 2,500 rpm s^{-1}), and as-formed films were heated at 120 °C for 10 min. The subsequent overlayers were deposited after the substrates cooled down to room temperature. Then, the PEAi solution (10 mg ml^{-1} in IPA) was spin-coated onto the PTAA-coated substrates at 5,000 rpm for 30 s (with a ramping rate of 2,500 rpm s^{-1}), and then the samples were heated at 100 °C for 5 min. Subsequently, the perovskite films were spin-coated onto the substrates by a two-step program at 2,000 rpm for 10 s (with a ramping rate of 200 rpm s^{-1}) and 6,000 rpm for 30 s (with a ramping rate of 1,000 rpm s^{-1}), respectively. During the second step, 100 μl of chlorobenzene was poured on the centre of the spinning substrates at 16 s before the end of the whole program. Then the samples were immediately transferred to the hotplate, annealing at 105 °C for 30 min.

For the NBG perovskite, the MeO-2PACz (0.5 mg ml^{-1} in ethanol) solution was spin-coated onto the pre-cleaned FTO substrates at 3,000 rpm for 30 s and annealed at 100 °C for 10 min in an N_2 -filled glove box. The perovskite films were prepared according to the procedure in a previous work³.

The samples were then transferred to a vacuum chamber to make the charge-selective layer without air exposure. C_{60} (35 nm) was thermally evaporated in a vacuum chamber with a base pressure of less than 4×10^{-4} Pa. For the NBG perovskites, the PCBM solution (20 mg ml^{-1} in chlorobenzene) was spin-coated on top of the perovskite layer at a speed of 1,000 rpm for 30 s (with a ramping rate of 200 rpm s^{-1}).

For the buffer layer fabrication, the samples were transferred to a separate vacuum chamber with a base pressure of less than 4×10^{-4} Pa. The YbO_x buffer layer with a thickness of 1.5 nm was obtained by the thermally evaporated metal Yb in the vacuum chamber. The thermal evaporation rate for the metal Yb is 1 $\text{\AA s}^{-1}$. Note that Yb will be naturally oxidized into the YbO_x in a few minutes without intentional additional oxidation processes. For the control devices with the BCP interlayer, BCP (1 mg ml^{-1} in ethanol) was spin-coated at 5,000 rpm with a ramping rate of 2,500 rpm s^{-1} for 30 s.

Finally, a 100-nm thick Cu electrode was thermally evaporated on the samples in the vacuum chamber (less than 4×10^{-4} Pa) through a metal shadow mask with an aperture area of 10.3 mm^2 . The aperture mask area for the photovoltaic performance tests is 7.2 mm^2 . Moreover, for the WBG and MBG devices, MgF_2 (100 nm) was deposited on the glass side of the ITO substrate as the anti-reflection coating, whereas for the NBG devices, MgF_2 (125 nm) was deposited on the glass side of the FTO substrate.

Stability assessment

For the stabilized power output test, the encapsulated devices were measured at the maximum power point (MPP) under full-spectrum white LED illumination (continuous 1-sun illumination) in ambient air or in the N_2 atmosphere. Thermal stability assessments for encapsulated devices were conducted by storing the encapsulated PSCs on a hot plate at 85 °C in an N_2 atmosphere. Photovoltaic performances were regularly measured when cooled down to room temperature. For the NBG-containing PSCs, the PCBM electron transport layer was replaced by C_{60} (35 nm, prepared by thermal evaporation) in the thermal stability and long-term MPP tracking assessments. The encapsulation

Article

for the device was done by capping the device with a glass slide, using UV adhesive as the sealant. The encapsulation process was finished in an N₂-filled glovebox (with H₂O and O₂ concentrations of less than 0.1 ppm). For the stability test under the ISOS-L-3 protocol, the temperature was controlled and monitored by the polyimide heating film with a digital display in temperature control. The ageing devices adhered to the polyimide heating film.

Photovoltaic characterization

The current density–voltage (J – V) characteristics of the PSCs were performed using a Keithley 2400 Source Meter under AM1.5 G illumination with a Xenon-lamp solar simulator (XES-40S1, SAN-EI). Before the measurement, the light intensity of 100 mW cm^{−2} was calibrated using a standard monocrystalline silicon solar cell with a KG-5 filter. The external quantum efficiency (EQE) was measured by a QE-R3011 measurement system (Enli Technology). Electroluminescence quantum efficiency (EQE_{EL}) was carried out at room temperature (297 K) in an N₂-filled glovebox. A Keithley 2400 Source Meter and an integration sphere (CME-LP38) coupled with a spectrometer (Flame, Ocean Optics) were used for the measurements. The device was placed on top of the integration sphere, and only forward light emission can be collected. The devices were swept from zero bias to forward bias.

Film characterization

The field-emission scanning electron microscope was used to investigate the top-view and cross-sectional morphology with an accelerating voltage of 10 kV (Hitachi U8010). Spin-coating perovskite films on the ITO-coated glass substrates prepared samples for ultraviolet–visible (UV–vis) absorption spectra characterizations. Then their UV–vis absorption spectra (350–850 nm) were recorded by a spectrophotometer (UH4150) with the background noise subtracted before testing. During the measuring process, the perovskite films were exposed to air for less than 5 min. The time-resolved photoluminescence spectroscopy measurements were performed by an Edinburgh Instruments FLS1000 photoluminescence spectrometer, and a pulsed laser measured all time-resolved photoluminescence-related tests with the excitation wavelength of 450 nm after 120-s illumination. Temperature-dependent J – V characteristics of ITO/[YbO_x (5 nm)/Cu (1 nm)]₂₀/Cu and ITO/[YbO_x (2 nm)]₂₀/Cu were measured using HP4140B picoammeter dominated by a computer within a cryostat. Both devices were processed by evaporating 20 alternative layers. For [YbO_x (5 nm)/Cu (1 nm)]₂₀, we mean we have deposited 20 alternating layers of 5 nm YbO_x followed by 1 nm Cu. The starting low temperature is 80 K cooled with liquid N₂, and data were acquired by every 30 K increase controlled by the MercuryITC low-temperature environment controller. Grazing-incidence wide-angle X-ray scattering measurements for amorphous Yb films deposited on various substrates were conducted at the BL17B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF). The X-ray wavelength was 1.240 Å with a grazing incident angle of 1.0°, and the scattering intensity was detected by a PILATUS 2 M detector.

Thermo Scientific ESCALAB Xi⁺ system was used to acquire the X-ray photoelectron spectroscopy, ultraviolet photoelectron spectroscopy and the reflection electron energy loss spectroscopy spectra. All samples were transferred into the analysis chamber using a vacuum transfer module avoiding exposure to the air. Unless otherwise stated, the gas cluster ion beam sputtering technology carried out the depth profiles with the parameters of 2,000 cluster sizes at 4 keV ion energy. Ar⁺ ion sputtering was used as a supplement to etch the hard materials, such as metal electrodes. High-resolution transmission electron microscopy and annular dark-field scanning transmission electron

microscopy images were obtained by using a JEM-2100F field-emission gun microscope with a 0.19 nm point-to-point resolution at 200 kV equipped with energy disperse X-ray detectors. The images were analysed using Gatan Digital Micrograph software. To reduce the electron dose during the high-resolution transmission electron microscopy characterization, we limited the total electron beam exposure time to within 5 s. All the time-of-flight secondary-ion mass spectrometry profiles were acquired by a PHI TRIFT V nanoTOF (ULVAC-PHI, Japan) system using a slice-and-view analysis scheme. A 20-kV C₆₀⁺ beam with a pulse frequency of 8,200 Hz and a pulse length of 15 ns was used as the primary-ion beam. All the secondary-ion signals were acquired from a 50 μm × 50 μm area. A 1-kV Ar⁺ beam was used as the sputter source to remove the surface material during time-of-flight secondary ions mass spectrometry profiling, and the sputtering crater size was 1,000 μm × 1,000 μm. All the secondary-ion data presented in the profiles were normalized by total ion intensities.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or the Supplementary Information and are also available from the corresponding authors on reasonable request.

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Author contributions D.L., Z.-H.L., P. Chen, Y.X. and R.Z. conceived the idea and designed the experiments. P. Chen, Y.X., S.L., X.J. and T.H. fabricated the PSCs and characterized the photovoltaic performances. P. Chen and M.Y. performed the electroluminescence tests. Y.X., P.K. and P. Caprioglio contributed to the analysis of electroluminescence results. X.L. conducted the EQE measurements. J.H., J.W., Y.S. and P.T. conducted the TEM and energy disperse X-ray tests. J.H. and N.C. performed REELS, UPS and XPS measurements. J.H., N.C., D.W., D.L., R.S. and Z.-H.L. contributed to analysing the REELS, UPS and XPS results. N.C. and Z.-H.L. contributed to the analysis of charge transport mechanisms. J.H., P.C., Y.X. and Q.L. conducted the scanning electron microscope measurements. S.L. and L.Z. helped perform the grazing-incidence wide-angle X-ray scattering measurements. P.C. and Y.X. measured the UV–Vis light absorption. C.-H.H., Y.-W.Y. and J.-J.S. performed the TOF-SIMS measurements. P.C., Y.X. and Q.Z. performed the time-resolved photoluminescence test and did data analysis. H.Y., Y.X. and P.C. conducted the stability test. D.L., Q.G., Z.-H.L., H.J.S. and R.Z. directed and supervised the project. P.C. and Y.X. wrote the first draft of the paper. D.L., Q.G., Z.-H.L., H.J.S. and R.Z. revised the paper. All authors analysed their data and reviewed and commented on the paper.

Competing interests Henry J. Snaith is the founder and Chief Scientific Officer of Oxford Photovoltaics, a company commercializing perovskite photovoltaics.

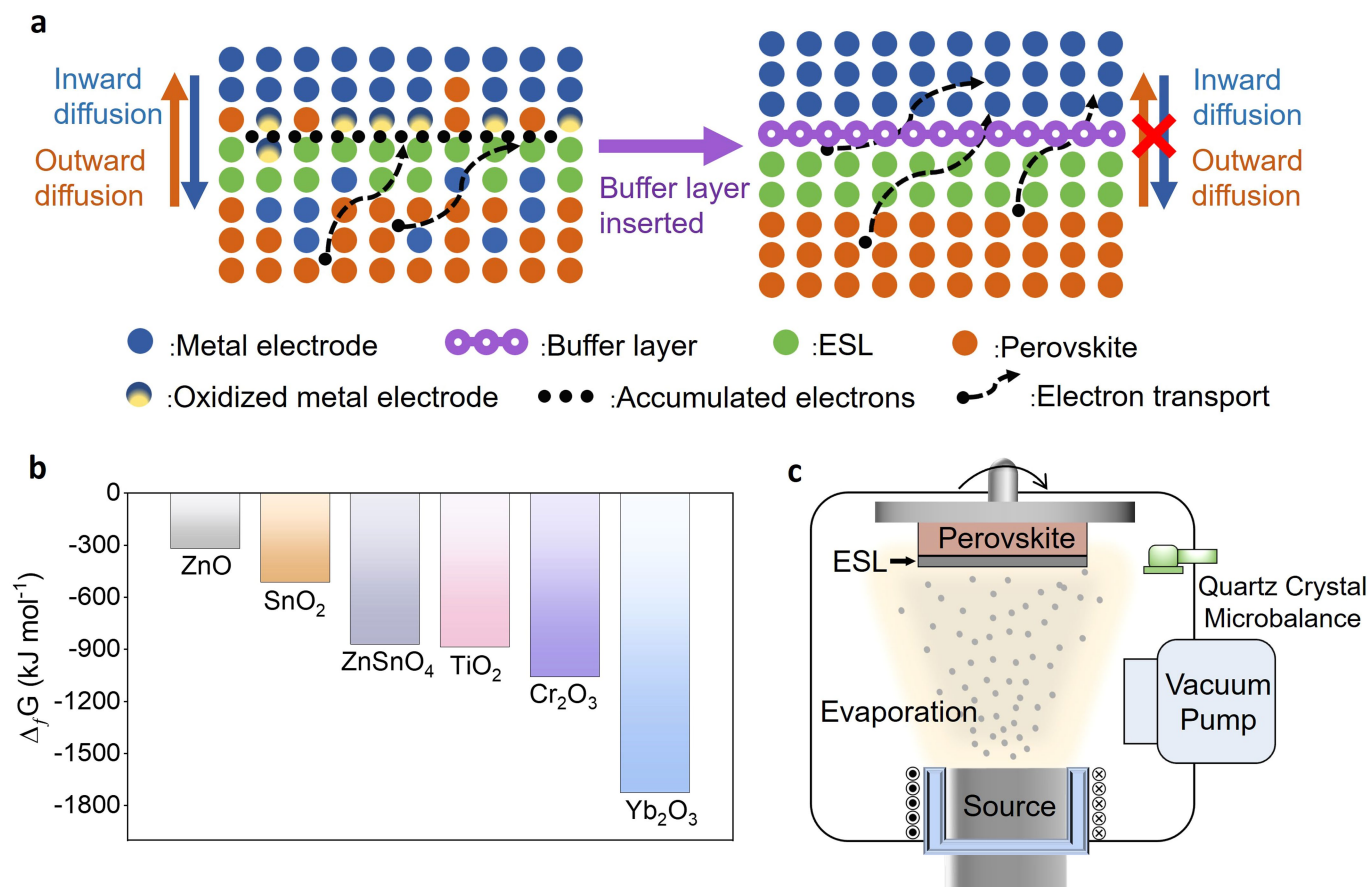
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-023-06892-x>.

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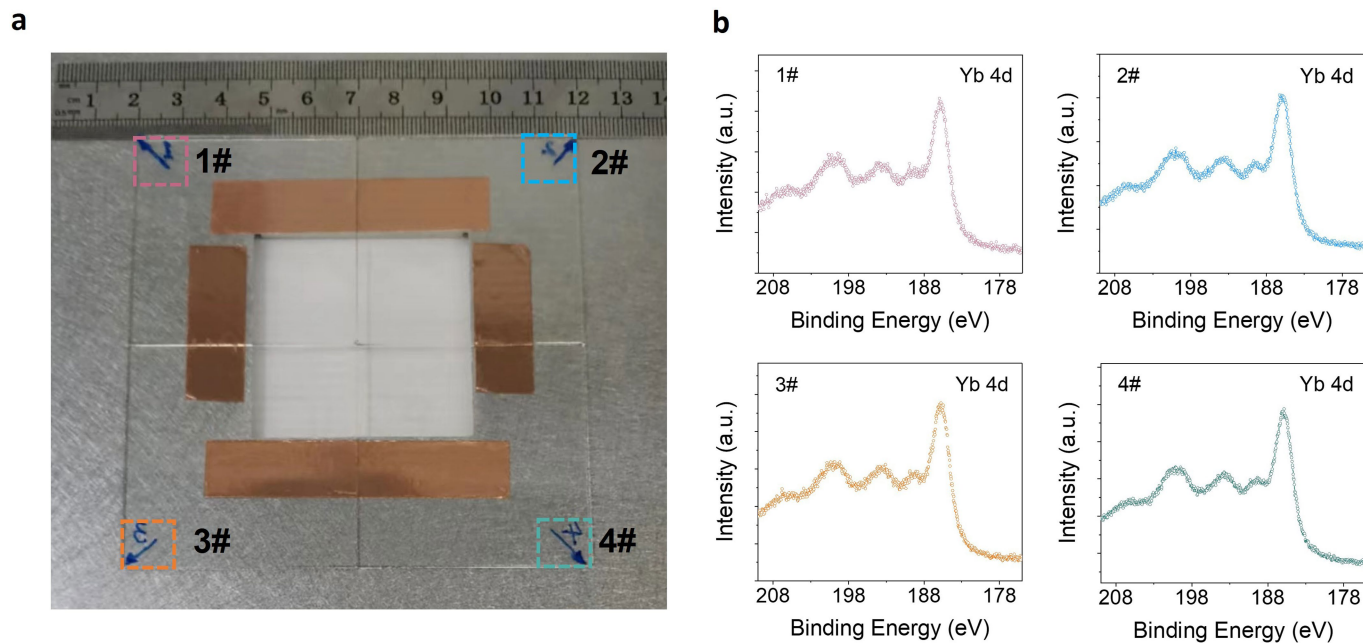
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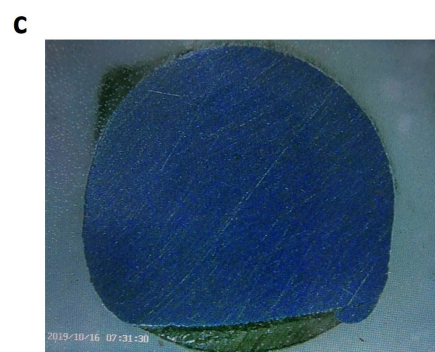
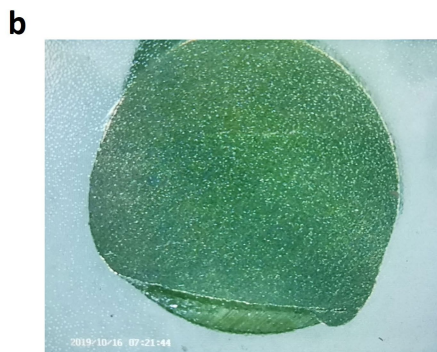
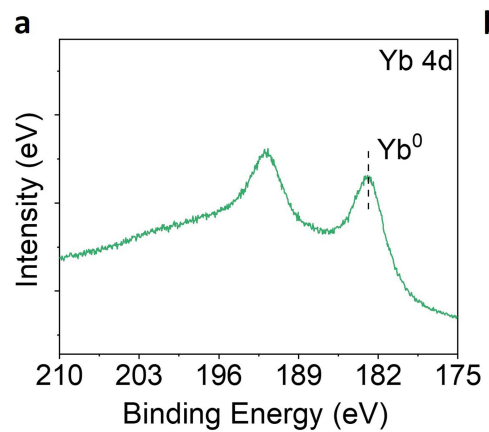
Extended Data Fig. 1 | Overview of buffer layers. **a**, Schematic illustration of the functions of an ideal buffer layer between the electron-selective layer (ESL) and the metal electrode, showing suppression of the harmful species movement and the non-radiative recombination. **b**, Standard molar Gibbs free energy ($\Delta_f G$)

of formation at 298.15 K in kJ mol⁻¹ of representative metal oxides including ZnO, SnO₂, ZnSnO₄, TiO₂, Cr₂O₃, and Yb₂O₃. **c**, Schematic illustration of the thermal evaporation deposition process of the metal Yb.



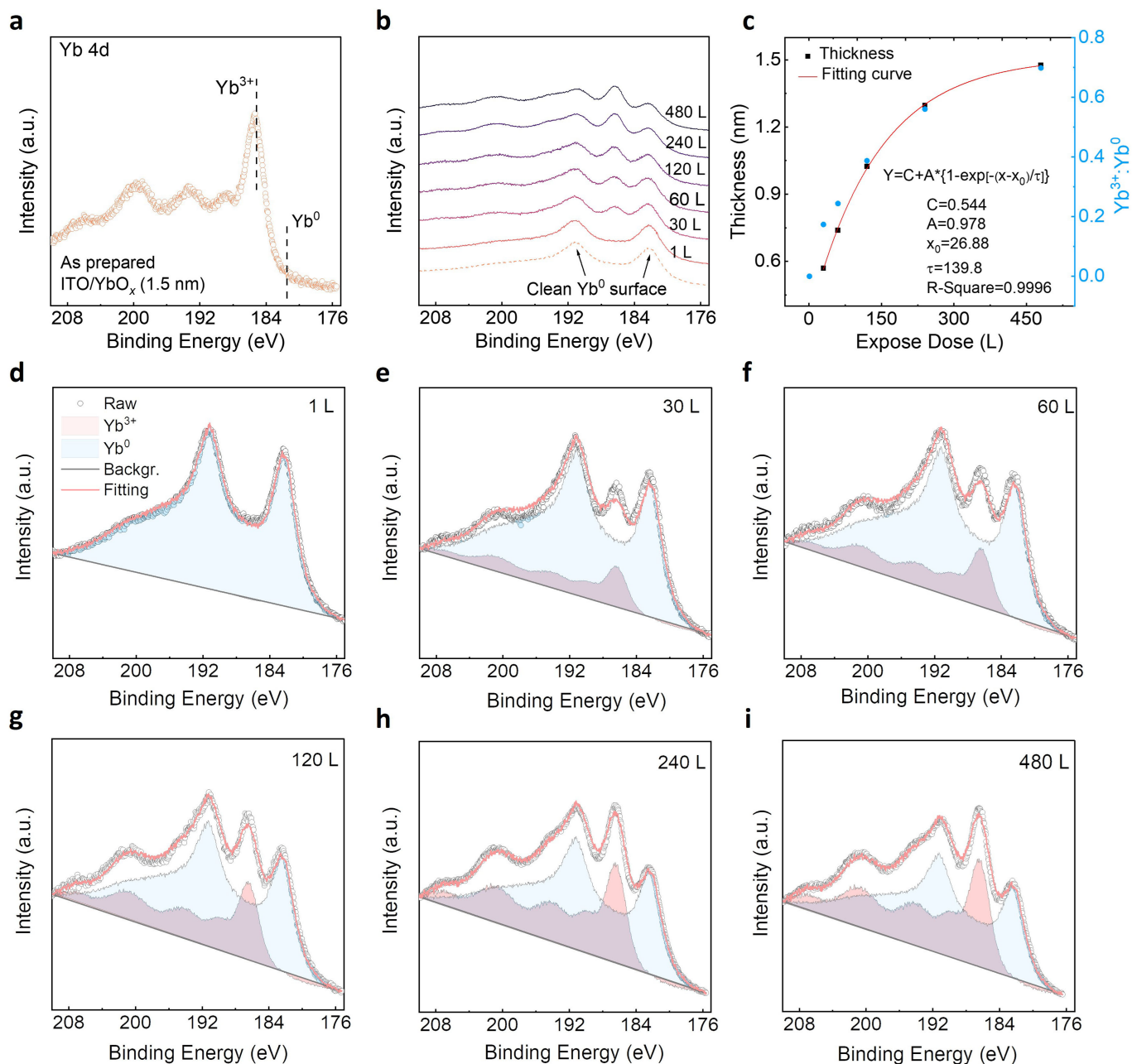
Extended Data Fig. 2 | XPS characterizations. **a**, Photograph of 1.5 nm YbO_x film deposited on a large-scale ITO substrate (100 cm^2) by a scalable thermal evaporation approach. The areas at the corner of the sample highlighted by the dotted box were used to conduct the XPS measurements. **b**, Yb 4d XPS spectra

corresponding to the highlighted areas in (a). The XPS data show that YbO_x on an 100-cm^2 substrate can be formed when upscaling in the device fabrication chamber with a base pressure of 10^{-4} Pa .



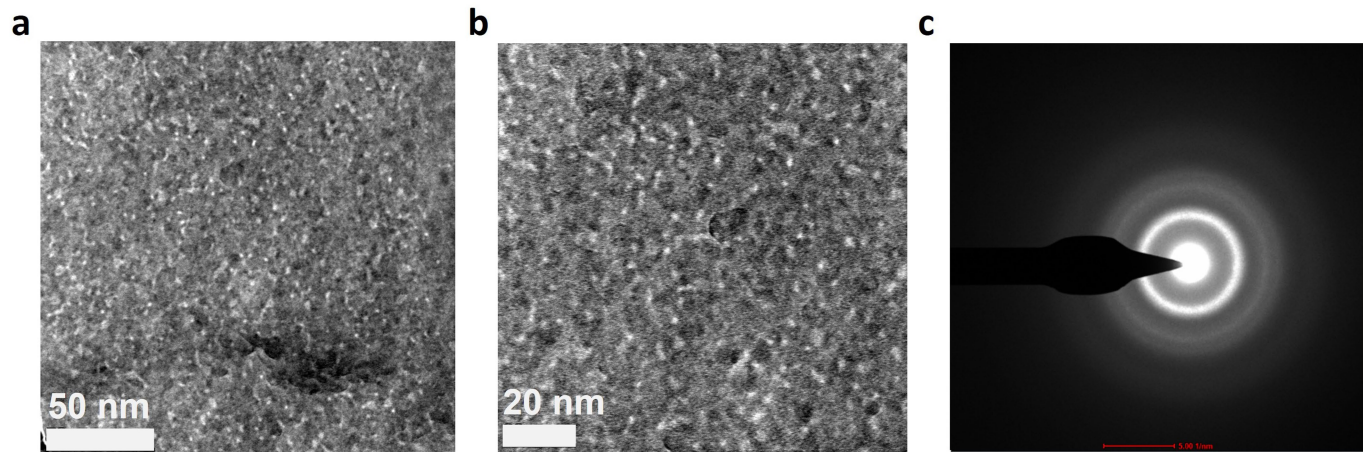
Extended Data Fig. 3 | XPS measurements and photographs of Yb bulk and YbO_x surface. a, Yb 4d XPS spectrum of the Yb bulk with 2400 s Ar ion etching. Photographs of the pristine Yb bulk with a naturally oxidized YbO_x surface (**b**)

and the polished Yb bulk (**c**) in ambient air. The colour change indicates that the refreshed metal Yb surface would quickly react with oxygen upon oxygen exposure, forming YbO_x on the surface.

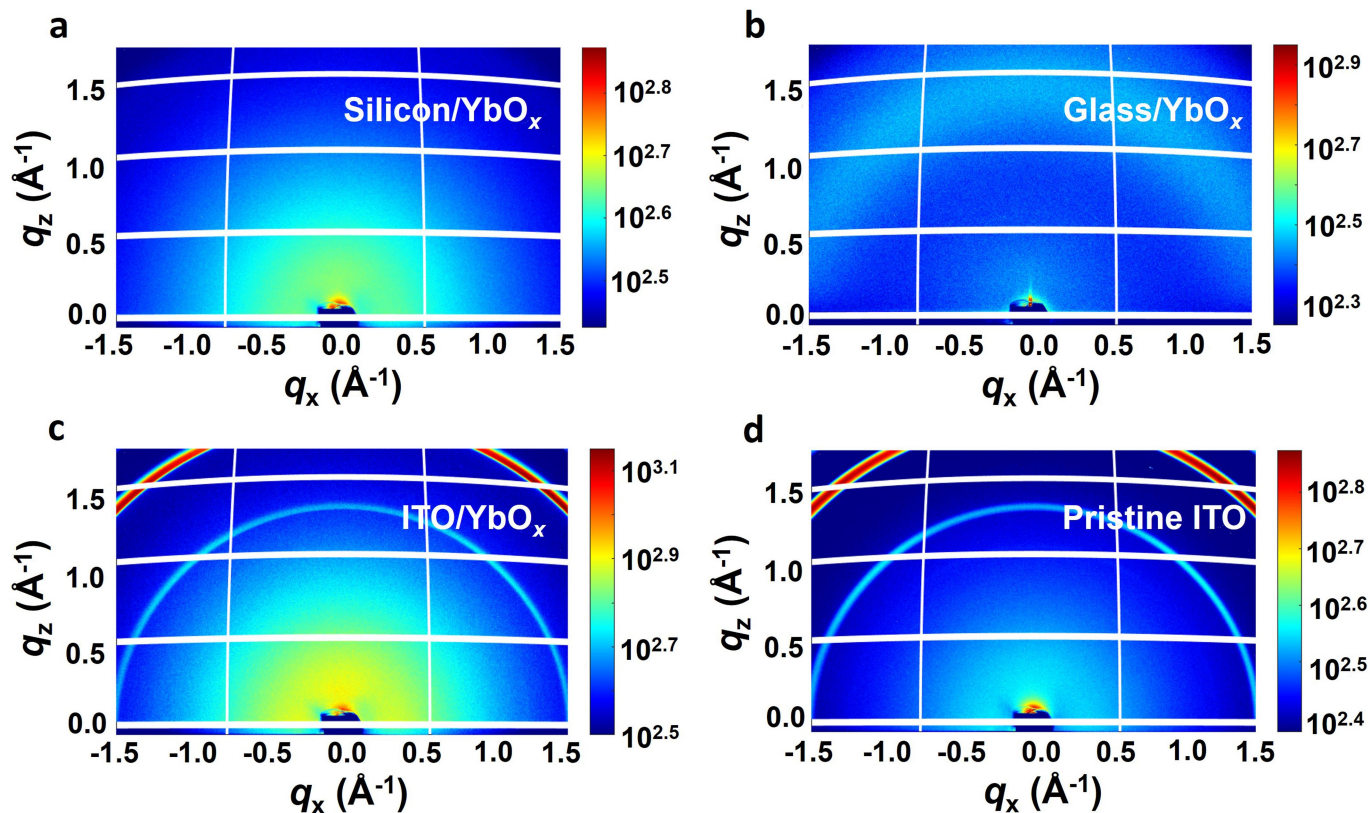


Extended Data Fig. 4 | The oxidation process of Yb. **a**, Yb 4d XPS spectrum of an as-prepared ITO/YbO_x (1.5 nm) film. The sample was immediately transferred from the physical vapour deposition system to the XPS chamber by the vacuum transfer module without exposure to the air. **b**, Yb 4d XPS spectra

of a clean Yb⁰ surface spontaneously oxidized in the UHV XPS chamber. **c**, The YbO_x thickness and Yb³⁺/Yb⁰ ratios as a function of the exposure dose. **d-i**, Yb 4d XPS spectra with increased exposure dose.

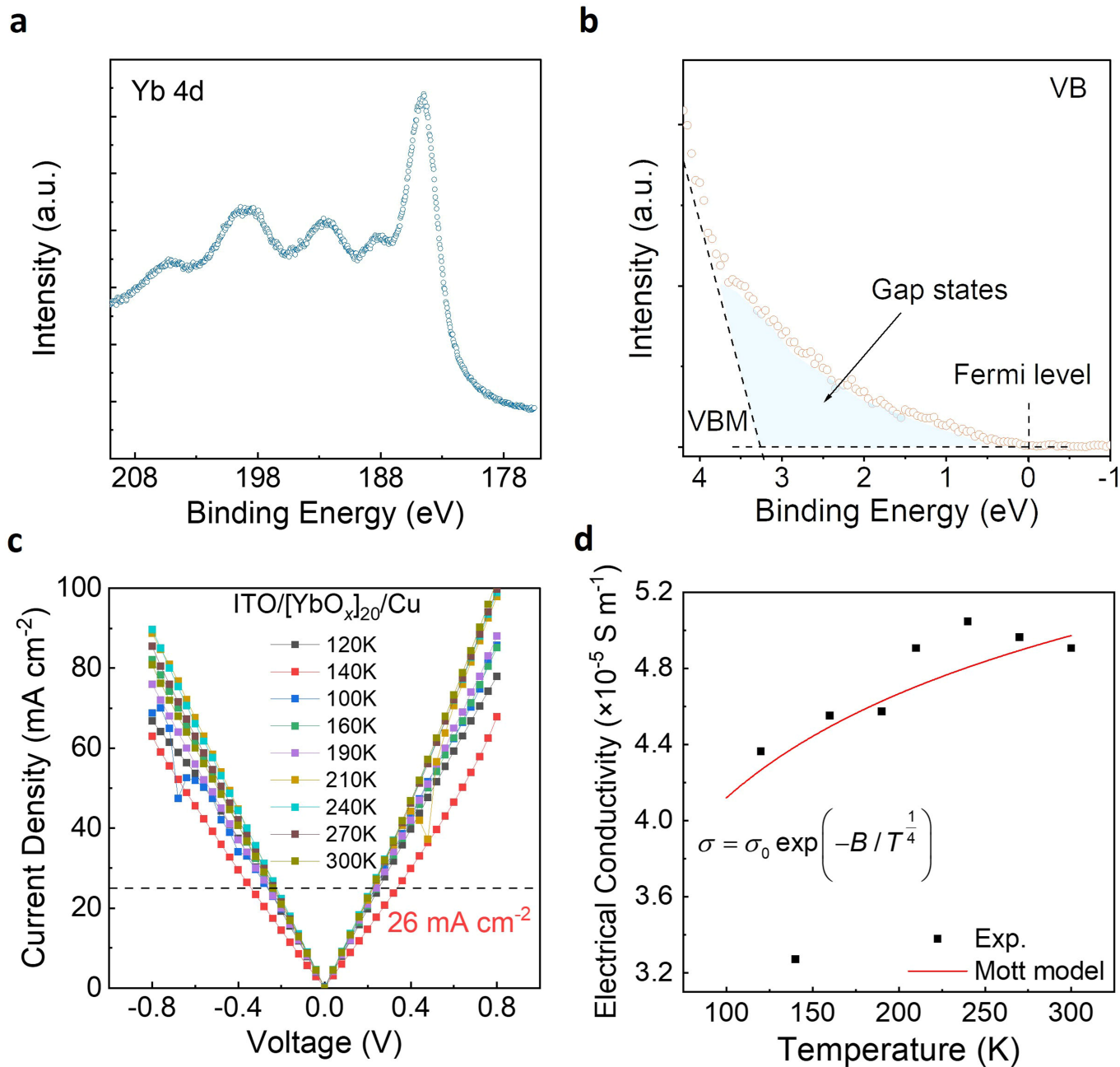


Extended Data Fig. 5 | Transmission electron microscopy (TEM) measurements. **a, b,** TEM images of YbO_x directly deposited on the carbon-supported copper (50 and 20 nm). **c,** The typical fast Fourier transform (FFT) spectrum ($30 \times 30 \text{ nm}^2$ area) indicates the amorphous crystal structure of YbO_x .



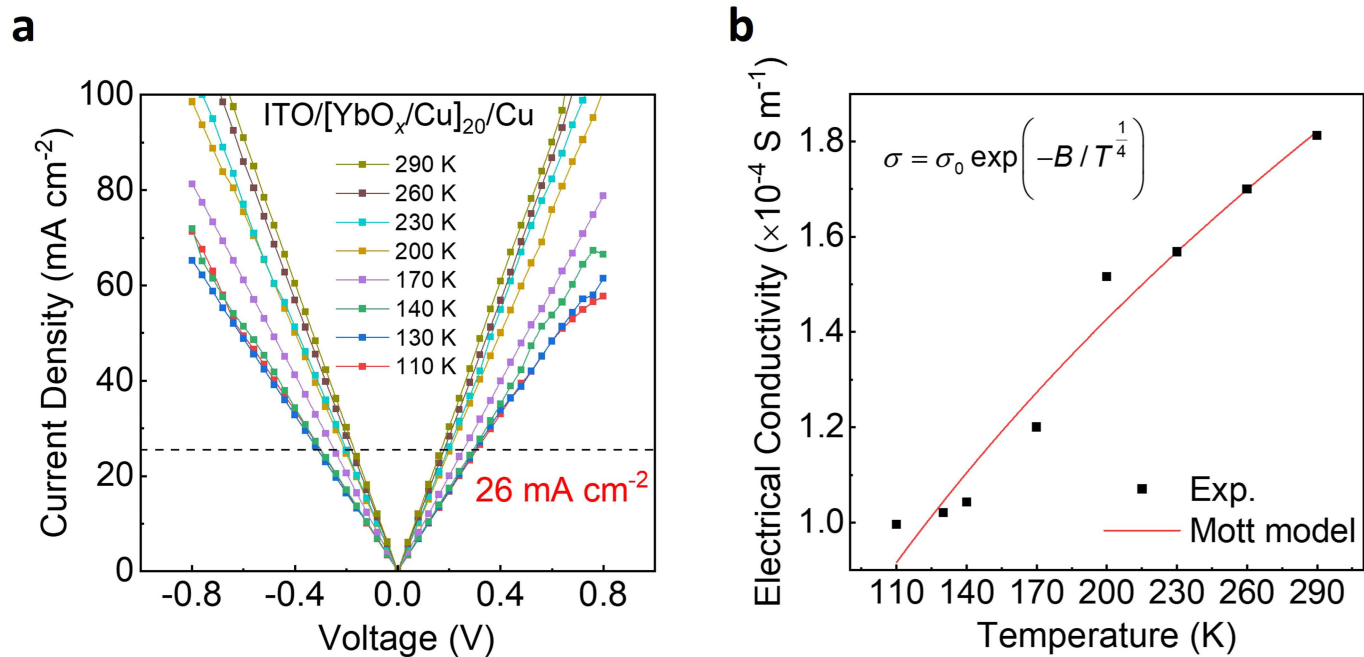
Extended Data Fig. 6 | 2D grazing incidence wide-angle X-ray scattering (GIWAXS) measurements. The 2D GIWAXS patterns at an incidence angle of 1.0° for YbO_x films prepared on different substrates including silicon (a), glass

(b), and ITO-coated glass (c), respectively. The thickness of the deposited YbO_x film for this measurement is 10 nm. d, The 2D GIWAXS pattern of the pristine ITO-coated glass.

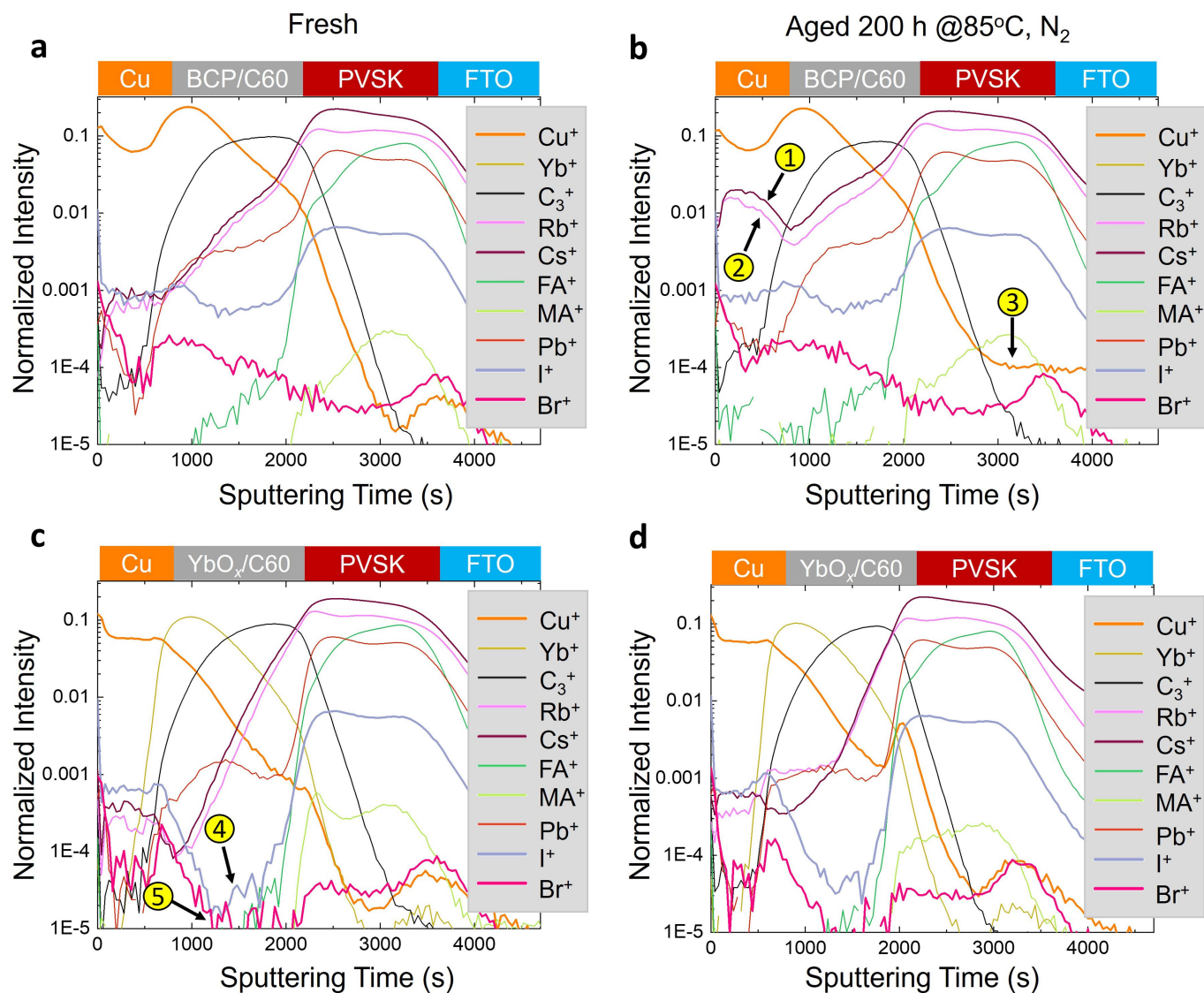


Extended Data Fig. 7 | Gap states in the YbO_x film and charge-carrier transport measurements. **a**, Yb 4d XPS spectra of the YbO_x film with a thickness of 40 nm. **b**, Valence band spectra of YbO_x. There exists a considerable density of gap states below the Fermi level. **c**, Current density-voltage characteristics of the ITO/[YbO_x (2 nm)]₂₀/Cu over temperature ranged from 120 K to 300 K. Note that the 40 nm YbO_x deposition is completed by evaporating 20 alternative

layers. **d**, The plot of conductivity against temperature. The conductivity is calculated at 26 mA cm⁻², where the injected current is equal to the short-circuit current of a narrow-bandgap PSC under 1-sun illumination. Note that the pre-factor $\sigma_0 = (2.22 \pm 0.47) \times 10^{-3} \text{ S m}^{-1}$, and the characteristic parameter $B = 10.33 \pm 0.81$.



Extended Data Fig. 8 | Charge-carrier transport measurements. **a**, Current density–voltage characteristics of the device with a configuration of ITO/[YbO_x (5 nm)/Cu (1 nm)]₂₀/Cu over temperature ranged from 110 K to 290 K. **b**, The plot of conductivity against temperature.



Extended Data Fig. 9 | Time-of-flight secondary ion mass spectrometry (TOF-SIMS) measurements. TOF-SIMS depth profiles of the BCP- or YbO_x-based PSCs with and without ageing. **a, c**, Control (with the BCP buffer layer) and target device (with the YbO_x buffer layer), respectively. **b, d**, TOF-SIMS profiles of both

control and target devices after ageing at 85 °C for 200 h in the N₂ atmosphere, respectively. The highlighted secondary ion distribution: 1: Cs⁺; 2: Rb⁺; 3: Cu⁺, 4: I⁺; 5: Br⁺.

Extended Data Table 1 | Photovoltaic parameters

Perovskite absorbers	Bandgap (eV)		V_{OC} (V)	J_{SC} (mA cm ⁻²)	FF	PCE (%)
FA _{0.83} CS _{0.17} Pb(I _{0.6} Br _{0.4}) ₃	1.77	Avg ^a	1.30 ± 0.01	18.5 ± 0.18	0.81 ± 0.01	19.7 ± 0.22
		Champ.	1.30	18.6	0.83	20.1
(CS _{0.05} FA _{0.85} MA _{0.10}) _{0.95} Pb(I _{0.85} Br _{0.15}) ₃	1.59	Avg ^a	1.17 ± 0.01	23.2 ± 0.18	0.80 ± 0.01	21.7 ± 0.28
		Champ.	1.18	23.1	0.81	22.1
Rb _{0.05} CS _{0.05} MA _{0.05} FA _{0.85} Pb(I _{0.95} Br _{0.05}) ₃	1.54	Avg ^a	1.16 ± 0.01	26.0 ± 0.14	0.82 ± 0.01	24.7 ± 0.28
		Champ.	1.16	26.1	0.83	25.2

Summary of photovoltaic parameters for the champion p-i-n PSCs with NBG (1.54 eV), MBG (1.59 eV), and WBG perovskites (1.77 eV). The devices were measured from reverse scans (RS) under simulated AM 1.5 G solar irradiation at 100 mW cm⁻². For p-i-n PSCs with WBG perovskites, RS is from 1.35 V to -0.02 V. For p-i-n PSCs with NBG and MBG perovskites, RS is from 1.22 V to -0.02 V.

^aThe average device parameters and standard deviations are based on 15 devices.

Solar Cells Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes	0.072 cm ²
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
Method used to determine the device area	☒ Yes	We employed a metal mask with 0.072 cm ² to define the device area.
	☐ No	<i>Explain why this information is not reported/not relevant.</i>

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	Supplementary Figure 9
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes	Both forward and reverse scans were applied in the hysteresis study. Dwell time was not applied during the measurements. The information is given in Methods - PV characterization
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes	All devices are measured in air at room temperature.
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
Protocol for preconditioning of the device before its characterization	☐ Yes	<i>State where this information can be found in the text.</i>
	☒ No	No preconditioning of the device before its characterization.
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes	Figure 3c-e, insert, Figure 4c, d
	☐ No	<i>Explain why this information is not reported/not relevant.</i>

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes	There is negligible hysteresis behavior observed for our devices.
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
Related experimental data	☒ Yes	Supplementary Figure 9
	☐ No	<i>Explain why this information is not reported/not relevant.</i>

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	Figure 3h
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	The integrated J _{sc} values from EQE were consistent with J _{sc} values from J-V measurements with mismatch ratio below 2%.
	☐ No	<i>Explain why this information is not reported/not relevant.</i>
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes	<i>State where this information can be found in the text.</i>
	☒ No	N.A.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

A 150 W class AAA solar simulator (XES-40S1, SAN-EI) was used as the simulated AM 1.5G standard light source for photovoltaic measurements, and the light intensity of our solar simulator for J-V measurements was calibrated by a standard monocrystalline silicon diode with a KG-5 filter.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

The light intensity was calibrated by a NIM calibrated standard monocrystalline silicon diode with a KG-5 filter before each measurement.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☐ Yes
☒ No

State where this information can be found in the text.

No spectral mismatch calculation was performed.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

0.072 cm²

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

State where this information can be found in the text.

No significant variations were observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

Measured by an independently accredited testing center (PHOTOVOLTAIC AND WIND POWER SYSTEMS QUALITY TEST CENTER (PWQTC), IEE, CHINESE ACADEMY OF SCIENCES), Supplementary Figure 10.

Explain why this information is not reported/not relevant.

A copy of any certificate(s)
Provide in Supplementary Information

☒ Yes
☐ No

Supplementary Figure 10

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

We tested 15 cells.

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

We show the statistics in Figure 3f and Supplementary Figures 11 and 12.

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

Figure 4a-d and Supplementary Figures 15 and 16

Explain why this information is not reported/not relevant.

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature